An aerial photograph taken on May 14, 2023, of PT. Indonesia Morowali Industrial Park (IMIP), one of the largest nickel producers in Sulawesi, Indonesia. Source: South China Morning Post (2024)

The (In)visible Cost of Indonesia's EV Ambitions:
Nickels, Human Rights, and Dirty Energy

Energy & Society – Zaidan Akbar

Indonesia electrifying dreams

Energy transition has been the main character in decarbonization for countries to reach their Net Zero Emissions goals. Changing the energy consumption pattern from fossil fuels to cleaner energy sources and transforming the country's power generation methods to produce electricity more sustainably are what the world is currently focusing on. These two approaches are intertwined and should be pursued together. Only focusing on one aspect, either upstream (power generation) or downstream (energy consumption), will not yield optimum results and can even hinder progress towards decarbonization. That's what's currently happening in Indonesia, where there is a big dream to electrify their transportation but yet to focus on their energy generation sources.

Indonesia showed their seriousness by setting a target that specified on their Government Regulation Number 55 of 2019 about Acceleration of Electric Vehicle Battery Programs for Road Transportation, that by 2030 they need to reach electric car (4W) sales of 2 million units and electric motorcycles (2W & 3W) sales of 13 million units. This number will not only come from importing EVs, but they also have set a strategy to produce EVs massively. (CORE, 2024).

The government committed to manufacture a total 9.6 million units EVs by 2030, with a target breakdown of 9 million electric motorcycles and 600 thousand electric cars. This target quantified in the Ministry of Industry Regulation No.6 of 2022. This target is sky high, looking at the current slow adoption rate in Indonesia with only 16 to 1 ratio of ICE compared to EV usage. While the market leader, China, has a ratio of 2:1 (Uchida et al., 2023).

What makes Indonesia firm about their goals and strategy is because their strategic advantage with the abundant nickel resources in hand. As the world's largest holders of nickel reserves, with 55 million metric tonnes, over half of the world's total nickel is located in Indonesia. Therefore, this offers as a promising opportunity to commence local production of nickel manganese cobalt (NMC) batteries, which can make EVs more affordable. Currently, nickel NMC are one of the most frequently used EV battery types (Yusgiantoro et al., 2021).

As global electric vehicle production continues to rise, Karabelli et al. (2020) predict that nickel demand for batteries in 2030 will account for 34% of the present mining capacity. This paper wants to highlight why this seemingly interesting market growth would bring lots of hidden cost that will not be visible by most people who benefitted by the electrification of vehicle. People who felt the benefit are dominated by urban population and only small fraction of these urban people can afford to buy EVs. This create a debate between expert in these energy transition narratives, whether this huge EVs goal is the best way to decarbonize if we reflect to the energy justice.

EV transition in Indonesia's current energy mix could drive (~~de~~)carbonization

In 2022, the IEA reported that Indonesia's emitted over 650 million metric tons of CO_2-equivalents from fuel combustion in the energy sector, with the transportation sector (23%) being the third-largest source of emissions after power generation (45%) and industry (25%). Within the transportation sector, road transport emerged as the main contributor, accounting for over 90% of emissions (Nabila and Aritenang, 2024). This might support the reason behind Indonesia focus to electrify their transports. But here's the harsh truth, Indonesia's national energy mix is still reliant by non-renewable energy, especially coal.

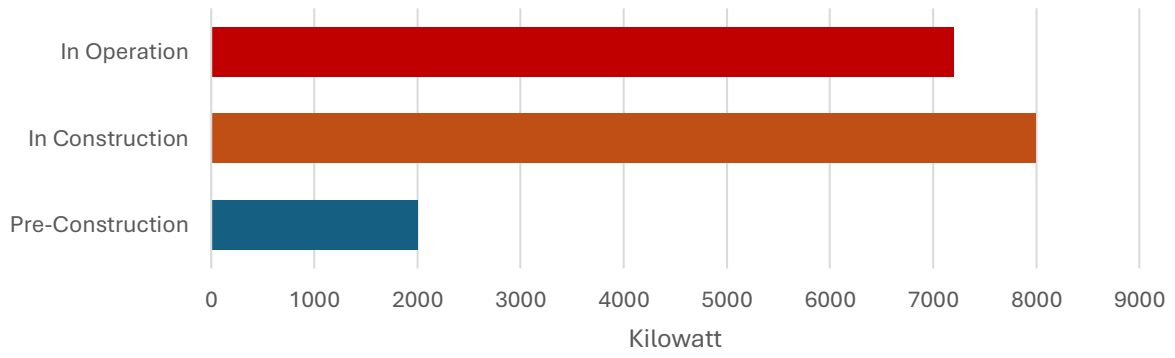
According to the National Energy Council (DEN), Indonesia's energy mix in 2023 was still heavily dominated by coal (41%), followed by petroleum (30%), natural gas (16%), and renewable energy (13%). This indicates that the electric vehicles (EVs) in the country will likely charge their electricity power sourced primarily from coal. While this may reduce reliance on petroleum, it is expected to increase dependence on other polluting energy sources, undermining the potential environmental benefits.

The environmental impact is further heightened by Indonesia's growing nickel production with government initiatives to support downstream projects. The number of operational nickel smelters as of January 2023 are 43, with 199 furnaces (Muliawati, 2023; Rahayu, 2023), consuming around 145 million tonnes of nickel ore annually. By 2025, it is projected that the number of smelters will rise to 136, with nickel ore demand rise significantly to 400 million tonnes per year. This surge in nickel processing contributes significantly to greenhouse gas emissions (Djamhari et al., 2024).

For every ton nickel produced in Indonesia, it generates an average of 58.6 tons of CO_2, 22% higher than the global average of 48 tons CO_2 per ton of nickel (Ho and Listiyorini, 2023). These emissions came from energy-intensive mining, transportation use in the industry, and the widespread use of coal power plants in nickel smelting. As a result, Indonesia's reliance on coal-based energy for nickel production not only offsets the potential benefits of EV adoption but also worsens its environmental challenges.

Indonesia reportedly operates captive coal-fired power plants dedicated to nickel smelters with a total capacity of 7.2 gigawatts (GW). Furthermore, an additional 8 GW are currently under construction, with 2 GW in the pre-construction phase (Global Energy Monitor et al., 2023).

Figure 1. Status and Capacity of Captive Coal-Fired Power Plants for Nickel Smelters in Indonesia, 2023



Source: Global Energy Monitor et al., 2023, prepared by the author.

This presents a difficult situation for EV manufacturers because they are facing pressure to address environmental, social, and governance (ESG) concerns along their supply chains, especially carbon emissions. Although some manufacturers prefer "low-carbon" nickel, its availability is not enough to meet projected demand and it often comes at a higher cost. This highlights the environmental risks associated with recent developments in Indonesia's nickel industry and the complex trade-offs involved in the decarbonization process (Castillo et al., 2022).

Zooming on the upstream, indigeneous communities have also raised complains about the environmental and health impact of nickel mining and processing. They highlight issues such as deforestation, water pollution, and the disruption of local livelihoods. These voices reveal the often-overlooked costs of the energy transition in Indonesia that highlight the need for deeper investigation and understanding.

Government actions towards achieving these dreams

With production rising, Indonesia is expected to supply over 60% of the world's nickel by 2030 (Lee, 2023). To support this goal, Indonesia's former president, Jokowi, has implemented a "commodity downstreaming" strategy to transform the country into an industrial powerhouse. Since 2020, Indonesia has banned raw nickel exports to promote local smelting and avoid the "resource curse." Major Chinese companies, like CATL and automakers such as Wuling Motors and BYD, have established manufacturing plants in Indonesia. Between 2014 and 2023, the country's nickel export value increased more than tenfold, driven by Chinese investment and technology transfer (Uchida et al., 2023). Additionally, the government has introduced various incentives to attract investors, manufacturers, and customers as can be seen in table 1 below.

Table 1. Indonesia's EV Incentives

Upstream	Corporate tax & tax allowance	• Corporate tax exemption for 5-20 years applies to investment values ranging from USD 7.2 million to USD 2.1 billion, plus an additional 2 years of tax reduction. • Up to 30% of the investment value can be deducted, calculated based on the valuation of tangible fixed assets, along with a 5% reduction in corporate income tax per year for 6 years (totaling 30%).
	Import duty on machinery	• Import duty exemption on capital goods (machinery) for two years during the construction phase. • Import duty exemption on goods and materials for production for two years during the initial production phase (or four years if locally produced machinery accounts for at least 30% of the total value).
	Capabilities building	• Gross income tax deduction of up to 300% for R&D expenses, 200% for costs associated with vocational training programs, and 60% for capital invested in fixed assets.
Downstream	Subsidy	• Subsidies of USD 5,130 are provided for locally manufactured four-wheel vehicles (4Ws) and USD 469 for electric two-wheelers (e2Ws) with at least 40% local content.
	Luxury & import tax	• Tax exemption for Battery Electric Vehicles (BEVs), previously taxed at 10%, compared to the 15%-40% tax rate for Internal Combustion Engine (ICE) vehicles. • Tax exemption also applies to Semi Knocked Down (SKD) imported BEVs with four or more wheels.
	Transfer & calculation tax	• A maximum tax ceiling of 10% is set, with specific rates determined by each provincial government.
	Preferential access	• Electric Vehicles (EVs) are exempt from the odd-even number plate traffic restrictions, applicable only in Jakarta.
Others	Charging infrastructure	• A ceiling rate of USD 0.16/kWh is set for charging service costs, with similar ceiling rates planned for fast and ultra-fast charging. • Subsidized EV charging is priced at USD 0.05/kWh.

Source: Indonesia Presidential Regulation No. 55 / 2019, prepared by the author.

Despite the various incentives in place, the adoption rate of EVs remains low in Indonesia. While many incentives such as tax exemptions, tax reductions and subsidies in the EV supply chain are commonly used in advanced countries, they may strain Indonesia's state budget due to the decreasing potential revenue and increasing government expenditures, especially for infrastructure development. In many cases, because of the government needing substantial funds to meet major development goals, they typically can gain more political support from their wide public impact initiatives like this EV ambition, yet it's crucial to focus on incentives that encourage EV adoption while ensuring efficient use of public funds.

Additionally, incentives that benefit wealthier individuals or corporations more that lower-income groups could widen socio-economic inequalities, as lower-income groups may struggle to afford EVs or fail to benefit equally from such policies. In the context of energy justice, it is crucial to ensure that the transition to EVs does not deepen these disparities. The government should focus on policies that prioritize equitable access to EV adoption and foster inclusive benefits. Additionally, promoting electric public transportation can help raise awareness, reduce reliance on private vehicles, and ensure that lower-income communities also gain the benefits of cleaner, more sustainable transportation options.

Energy Justice considerations

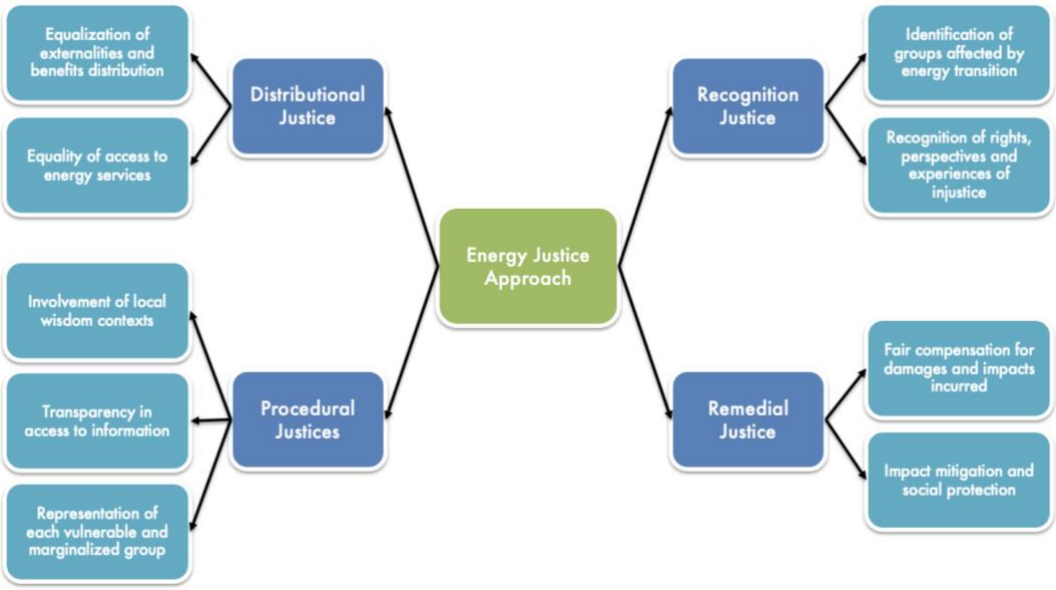
Energy transition always comes as a two-edged sword, a noble purpose but with many perspectives to reflect. Improvement in one side might result in a degradation in another. That's why finding sustainable energy solutions that can distributes the benefits and cost fairly and reasonably is a must in a representative energy decision-making system. Things need to be evaluated in the whole energy systems, not only focusing on upstream and downstream. (Sovacool et al., 2017; Sovacool & Dworkin, 2015).

The energy system refers to the entire process of getting energy, from extracting raw materials (in this case nickel mining) to producing, transmitting, and distributing it, and finally, how it's consumed and disposed of. (Bevier, 2008; Alanne and Saari, 2006; Gagnon et al. 2002). The system is not just about physical infrastructure (like power plants or grids), but also includes the social aspect, meaning how technology is connected to people's behaviors, lifestyles, values, and organizations (Markard et al., 2012; Whitmarsh, 2012; Kern and Smith, 2008).

However, in Indonesia, they overlook the justice issues in the entire system, such as the environmental damage caused by nickel mining for EV batteries and the unfair EV incentives distribution. While most energy sources don't fully consider the environmental and social costs before and after their use, electric vehicles are no exception.

In the larger context of energy justice, these issues challenge the four energy justice frameworks by Jenkins et al. (2016), and Heffron and McCauley (2017), which were adapted from environmental justice research. These frameworks consist of recognitional justice, distributional justice, procedural justice, and remedial justice, which all are important for examining inequities in energy policy.

Figure 2. Energy Justice Approaches



Source: (Heffron & McCauley, 2017; Jenkins et al., 2016; McCauley et al., 2019; Sovacool et al., 2017) prepared by the author.

The first lens, distributional justice, is concerned with the fairness of who receives what benefit. It relates to the equitable distribution of both the costs and benefits associated with the energy transition process (Lee and Byrne, 2019). This lens of justice underlines equal access to energy and its benefits for all members of society, regardless of their social, economic, or geographic background. In the case of nickel mining in Indonesia, the main beneficiaries are the foreign mining companies, followed by the government through the collection of taxes.

On the other part, local communities that suffered from negative impacts of the mines and smelters, rarely received fair benefits. They don't get to experience the energy transition firsthand, like using EVs for transportation, which is more accessible to urban populations. Distributional justice does not only concern about where the infrastructure is built, but also making sure people have access to energy services and considering the risks of processing energy resources in the future (Jenkins et al., 2016; McCauley et al., 2019).

The second lens, recognition justice, looks to answer the question about "Are the rights and perspectives of every stakeholder respected?". This lens highlights the importance of acknowledging the interests and rights of all individuals and groups, especially those who are vulnerable and marginalized during the energy transition process (Lee and Byrne, 2019). It is important to reflect on which groups are most impacted by energy transition (McCauley et al., 2019).

What this means is to involve the needs and aspirations of vulnerable groups in society, such as workers, indigenous communities, women, and other marginalized populations. Their perspectives are essential in decision-making consideration to achieve justice. Therefore, recognition justice highlights the importance of valuing the experiences of injustice faced by marginalized groups as a basis for policy decisions, particularly in ensuring a fair energy transition (Jenkins et al., 2016).

From the third lens, procedural justice, involves addressing the question of “Do local communities have real involvement in this project?” while exploring how decision-makers have engaged with these communities. In other words, procedural justice is focuses on applying fair processes and ensuring the inclusive participation of all stakeholders in a non-discriminatory way (Jenkins et al., 2016). It underscores the importance of open, participatory, and transparent decision-making within energy policy to ensure that energy decisions align with the interests and needs of the entire society.

Related to this, Jenkins et al. (2016) introduce three inclusion mechanisms to promote justice, which consist of incorporating local knowledge, ensuring transparent access to information, and providing representative institutional participation. Jenkins highlights that local knowledge plays a key role in enhancing inclusiveness and public engagement, particularly for affected groups such as indigenous communities. In the nickel industry, meaningful participation goes beyond physical involvement to include the integration of local wisdom, narratives, and traditions of indigenous peoples as critical considerations, ensuring they have a substantial influence on policy decisions.

Last but not least, remedial justice, at the end if everything already been done, we are trying to solve the question of “Are the companies fixing past harms and making people accountable?”. This approach seeks to ensure fair compensation for those impacted by energy transition. Heffron and McCauley (2017) explain that remedial justice helps identify preventive actions and encourages communities and stakeholders to confront and address forms of injustices or harm. By incorporating remedial justice into decision-making, policymakers are urged to consider the potential damages and injustices that might result from energy policies.

Although some communities receive "compensation," it is often not distributed fairly among all those affected. Financial agreements are typically made to offset inconveniences caused by mining activities and the heavy vehicles that pass through their living areas. However, some view these payments as a form of bribery to keep the mining operations going.

EVs are not a One-Size-Fits-All solution

In conclusion, the EV energy transition has widespread global impacts. While demand and policy may originate from the global north, the supply and execution often depends on countries in the global south, where unjust practices may be taking place. Policies implemented in the global north not only affect local communities but also trigger a ripple effect throughout global energy systems. As Holifield et al. (2009) highlight, "policies and practices specific to one place can have impacts across national borders, affect multiple levels, and spread across global networks”.

And to make it clear, adopting electric vehicles does not automatically lead to decarbonization. Various studies have shown that the environmental impact, good or bad, of the EV depends on the energy mix used for electricity generation and the source carbon emission intensity (Karplus et al., 2009). Hawkins et al. (2013) discovered that while electric vehicles powered by coal-based electricity could reduce emissions like PM2.5, they may actually lead to higher CO_2 emissions. Similarly, ERIA’s (2020) and Doucette and McCulloch (2011) research’s results suggest that countries such as India, China, and Indonesia will not gain the benefits of electric vehicle unless they also transition to low-carbon power generation. Therefore, for electric vehicles to truly contribute to decarbonization, a simultaneous shift toward cleaner energy sources is essential.

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P.S. *AI is used for grammar correction*.